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Toward a transport-based analysis of nutrient spiraling and uptake in streams

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Abstract

Nutrient addition experiments are designed to study the cycling of nutrients in stream ecosystems where hydrologic and nonhydrologic processes determine nutrient fate. Because of the importance of hydrologic processes in stream ecosystems, a conceptual model known as nutrient spiraling is frequently employed. A central part of the nutrient spiraling approach is the determination of uptake length (S_w), the average distance traveled by dissolved nutrients in the water column before uptake. Although the nutrient spiraling concept has been an invaluable tool in stream ecology, the current practice of estimating uptake length from steady-state nutrient data using linear regression (called here the "S_w approach") presents a number of limitations. These limitations are identified by comparing the exponential S_w equation with analytical solutions of a stream solute transport model. This comparison indicates that (1) S_w is an aggregate measure of uptake that does not distinguish between main channel and storage zone processes, (2) S_w is an integrated measure of numerous hydrologic and nonhydrologic processes—this process integration may lead to difficulties in interpretation when comparing estimates of S_w, and (3) estimates of uptake velocity and areal uptake rate (v_f and U) based on S_w are not independent of system hydrology. Given these findings, a transport-based approach to nutrient spiraling is presented for steady-state and time-series data sets. The transport-based approach for time-series data sets is suggested for future research on nutrient uptake as it provides a number of benefits, including the ability to (1) separately quantify main channel and storage zone uptake, (2) quantify specific hydrologic and nonhydrologic processes using various model parameters (process separation), (3) estimate uptake velocities and areal uptake rates that are independent of hydrologic effects, and (4) use short-term, non-plateau nutrient additions such that the effects of regeneration and mineralization are minimized. In summary, the transport-based, time-series approach provides a means of estimating traditional measures of nutrient uptake (S_w, v_f, U) while providing additional information on the location and magnitude of uptake (main channel versus storage zone). Application of the transport-based approach to time-series data from Green Creek, Antarctica, indicates that the bulk of nitrate uptake (~74% to 100%) occurred within the main channel where benthic uptake by algal mats is a likely process. Substantial uptake (~26%) also occurred in the storage zone of one reach, where uptake is attributed to the microbial community.

Introduction

"Differentiation between hydrologic and nonhydrologic processes is critical to an understanding of solute dynamics."
—Stream Solute Workshop (1990)

Many studies of nutrient cycling in streams have focused on small watersheds in which the quantification of nutrient fluxes and the development of nutrient budgets is scientifically and logistically tractable (e.g., Meyer 1979, Grimm et al. 1981, Triska et al. 1984, Likens and Bormann 1995). This emphasis on small watersheds is appropriate given recent research highlighting the importance of small streams in processing and retaining nutrients (Alexander et al. 2000, Peterson et al. 2001). A key component of many small watershed studies has been the experimental addition of nutrients (Meyer 1979, Mulholland et al. 1983, Triska et al. 1989, Stream Solute Workshop 1990). Nutrient addition experiments are designed to study the cycling of nutrients in stream ecosystems where hydrologic and nonhydrologic processes determine nutrient fate. Because of the importance of hydrologic processes in stream ecosystems, a conceptual model of nutri-

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ent cycling known as nutrient spiraling has been developed (Webster and Patten 1979, Newbold et al. 1981). Under the conceptual model, a nutrient atom within the water column is transported downstream in dissolved form and is later removed from the water column by biotic processes (e.g., uptake by periphyton). The nutrient spiral is completed when the nutrient atom returns to the water column after release from the biotic compartment. A quantitative framework for the study of nutrient spiraling has been developed in which spiraling length is defined as the sum of uptake length (S_w) and turnover length (S_b). Under this definition, spiraling length represents the distance an average nutrient atom travels downstream in dissolved form (S_w) and within the biotic compartment (S_b). Analysis of nutrient cycling using the nutrient spiraling concept has been an invaluable tool in stream ecology, as it provides a framework for quantifying nutrient use in diverse stream ecosystems.

Despite some recent advances in the study of nutrient spiraling (Mulholland et al. 2002, Thomas et al. 2003, Payn et al. 2005, Doyle 2005), the basic approach has been essentially unchanged over the last two decades. In this article, a critical analysis of the spiraling approach is presented, with an eye toward updating the approach to incorporate contemporary advances in modeling the dynamics of ecosystem processes. One of these advances is the widespread availability of computer software for the analysis of stream transport (e.g., OTIS; Runkel 1998). Models of stream transport provide another approach to the study of nutrient uptake (Kim et al. 1990, 1992, Tate et al. 1995, Gooseff et al. 2004, Gücker and Boëchat 2004, McKnight et al. 2004) that has been used much less frequently than the spiraling concept.

This article begins with a description of the spiraling and transport modeling approaches, and proceeds to show how the two seemingly disparate approaches are fundamentally linked. This linkage is based on a comparison of the equation used to estimate uptake length (S_w) from steady-state data and analytical solutions of a stream transport model. This comparison indicates that (1) S_w is an aggregate measure of uptake that does not distinguish between main channel and storage zone processes, (2) S_w is an integrated measure of numerous hydrologic and nonhydrologic processes—this process integration may lead to difficulties in interpretation when comparing estimates of S_w, and (3) estimates of uptake velocity and areal uptake rate (v_f and U) based on S_w are not independent of system hydrology. Given these findings, a new transport-based approach is presented for steady-state and time-variable data sets. The transport-based approach for time-series data sets is suggested for future research, as it directly addresses the limitations associated with S_w.

Existing approaches for modeling uptake

Experimental additions of nutrients typically involve a constant-rate injection of a nutrient solution that is coinjected with a conservative tracer at the upstream end of the study reach. Studies of nutrient uptake use the temporal and/or spatial changes in the concentrations of the added nutrients to quantify nutrient cycling. Two types of data may be collected during the experimental addition: *time-series data* are collected at one or more locations along the study reach by collecting water samples before, during, and after the injection; *steady-state data* are collected at several locations along the study reach after the conservative tracer has reached a steady-state plateau. Two approaches to analyzing data from experimental additions of nutrients are described in the subsections that follow. A third subsection describes how the two approaches are mathematically linked.

Nutrient spiraling—Under the nutrient spiraling concept, the average distance traveled by a nutrient atom while completing a nutrient spiral is given by:

$$S = S_w + S_b \tag{1}$$

where S is spiraling length, S_w is uptake length, and S_b is turnover length. Uptake length (S_w) is defined as the average distance traveled by dissolved nutrients in the water column before uptake, whereas turnover length represents the downstream movement of nutrients in the biotic compartment.

Techniques and methods for estimating spiraling length from experimental data are presented in the landmark papers of Newbold et al. (1981, 1983). Both of these papers describe the analysis of a data set obtained from a radiolabeled phosphate addition (^{32}P). Use of radiolabeled phosphate allowed the investigators to track the movement of phosphate through multiple biotic compartments, thereby subdividing S_b into its multiple components (Newbold et al. 1983). In regard to S_w, Newbold et al. (1981) present a linear regression technique in which steady-state data showing longitudinal declines in phosphate concentration are used to estimate uptake length. In the 1983 paper, in contrast, the authors determine uptake length from time-series data, using a transport modeling approach and mass balance calculations. This latter paper provides a detailed phosphate budget and illustrates the power of transport modeling to aid in process separation and quantification (Newbold et al. 1983). Despite this illustration, the bulk of nutrient spiraling applications to date have used the steady-state approach to estimate S_w by linear regression (Mulholland et al. 1983, Munn and Meyer 1990, Valett et al. 1996, Marti et al. 1997, Butturini and Sabater 1998, Davis and Minshall 1999, Tank et al. 2000, Haggard et al. 2001, Wollheim et al. 2001, Dodds et al. 2002, Hall et al. 2002). Estimation of S_w using steady-state data and linear regression, called here the "S_w approach," is therefore the focus of this article.

A theoretical basis for the S_w approach was presented by the Stream Solute Workshop (1990). In short, the one-dimensional advection-dispersion equation with first-order uptake may be solved analytically for the case of steady-state conditions and negligible longitudinal dispersion. This solution yields

$$C_x\left(\frac{Q_x}{Q_0}\right) = C_0 e^{-\frac{1}{S_w}x} \tag{2}$$

where C is nutrient concentration (ML^{-3}), S_w is the uptake length (L), Q is stream discharge (L^3T^{-1}), x is distance (L), and

the subscripts "0" and "x" denote quantities at the injection site and at downstream locations, respectively (units of mass [M], length [L], and time [T] are used throughout this section). Nutrient concentrations (C_0 and C_x) are corrected for background by subtracting out concentration data obtained before the experimental addition. Because Eq. 2 is a function of uptake length, it will be referred to herein as the "S_w equation." Applications of the S_w approach typically use the linear form of Eq. 2:

$$\ln\left(C_x\frac{Q_x}{Q_0}\right) = \ln(C_0) - \frac{1}{S_w}x \qquad (3)$$

where $1/S_w$ and $\ln(C_0)$ are the slope and intercept of a linear regression relating dilution-corrected concentration (C_xQ_x/Q_0) and distance. Given the linear regression estimate of S_w, uptake velocities (v_f) and areal uptake rates (U) may be calculated (Stream Solute Workshop 1990, Davis and Minshall 1999, Hall et al. 2002).

Widespread application of the S_w approach may be attributed to two factors. First, detailed studies such as that of Newbold et al. (1983) have indicated that uptake length accounts for most of the spiraling length ($S_w >> S_b$), such that S_w is primary quantity of interest. Second, calculation of S_w using steady-state data and linear regression requires a relatively simple data collection and analysis effort. As a result, numerous estimates of S_w have appeared in the literature, representing a wide variety of stream ecosystems. Because uptake length is defined as the distance traveled prior to uptake, S_w is a function of both hydrologic (advection, dispersion, inflow, transient storage) and nonhydrologic (uptake rates, biomass standing stock, temperature) factors. An advantage of the S_w approach is therefore the ability to integrate the suite of hydrologic and nonhydrologic factors that affect downstream movement of nutrients into a single metric, S_w. As will be shown later in this article, this advantage of the S_w approach also represents a disadvantage when individual processes are of interest.

Transport modeling—Models of stream transport vary in complexity, from fully three-dimensional models with complex hydrologic and biogeochemical interactions to simple one-dimensional box models that ignore the effects of mixing (dispersion and transient storage). As noted by Newbold et al. (1983), analysis of uptake data using complex models requires extensive data collection efforts that may not be within the scope of many studies. At the other extreme, use of simple models (e.g., the S_w approach; Eq. 2) may limit the investigator's ability to differentiate between hydrologic and nonhydrologic processes, a key aspect of understanding nutrient dynamics (Stream Solute Workshop 1990). A compromise between these two extremes is the use of the one-dimensional transient storage model (Bencala and Walters 1983, Runkel 1998). The transient storage model is based on the advection-dispersion equation, with additional terms to account for transient storage, lateral inflow, and first-order uptake. Two conceptual areas are defined within the model: the main channel, defined as the portion of the stream in which advection and dispersion are the dominant transport mechanisms, and the storage zone, defined as the portion of the stream that contributes to transient storage (i.e., the hyporheic zone, pools, and eddies). Exchange of solute mass between the main channel and the storage zone is modeled as a first-order mass transfer process. Given this conceptual framework, equations describing the spatial and temporal variation in solute concentrations in the main channel and transient storage zone are given by

$$\frac{\partial C}{\partial t} = -\frac{Q}{A}\frac{\partial C}{\partial x} + \frac{1}{A}\frac{\partial}{\partial x}\left(AD\frac{\partial C}{\partial x}\right) + \frac{q_L}{A}(C_L - C) + \alpha(C_S - C) - \lambda C \qquad (4)$$

$$\frac{dC_S}{dt} = \alpha\frac{A}{A_S}(C - C_S) - \lambda_S C_S \qquad (5)$$

where A is the main channel cross–sectional area (L^2), A_S is the cross-sectional area of the storage zone (L^2), C is the main channel concentration (ML^{-3}), C_S is the storage zone solute concentration (ML^{-3}), C_L is the lateral inflow solute concentration (ML^{-3}), D is the dispersion coefficient (L^2T^{-1}), q_L is the lateral inflow rate on a per length basis ($L^3T^{-1}L^{-1}$), t is time (T), α is the storage zone exchange coefficient (T^{-1}), λ is the main channel first-order uptake coefficient (T^{-1}), and λ_S is the storage zone first-order uptake coefficient (T^{-1}).

Under the transient storage framework (Eq. 4 and Eq. 5), the effects of hydrologic processes are represented by a suite of physical parameters (A, A_S, D, Q, q_L, and α), whereas nonhydrologic processes (biotic uptake and abiotic sorption) are modeled using first-order uptake coefficients (λ, λ_S). A typical application of the transport modeling approach proceeds in two steps: physical parameters are first estimated using data from the conservative tracer; given this description of the hydrologic system, first-order uptake coefficients are estimated using data from the nutrient addition. Both of these steps have been facilitated through the use of nonlinear regression techniques for parameter estimation (Wagner and Gorelick 1986, Runkel 1998), as illustrated in McKnight et al. (2004). This separation of hydrologic and nonhydrologic processes as a two-step simulation exercise is a key aspect of understanding stream dynamics (Stream Solute Workshop 1990) that takes full advantage of the available data (i.e., the co-injection of the nutrient solution and conservative tracer).

Linking the S_w and transport-based approaches—The relationship between the S_w and transport-based approaches to modeling uptake may be seen by developing analytical solutions to the transport equations (Eq. 4 and Eq. 5) and comparing those solutions to the S_w equation (Eq. 2). This comparison is presented for four cases of increasing complexity (Table 1). Case I is provided to illustrate the basis of the S_w approach, whereas Cases II to IV illustrate the complexities of natural systems and the implications for various spiraling metrics. In the development of analytical solutions that follows, three simplifying assumptions are used for all of the

Table 1. Summary of Cases I to IV.

Case	Processes considered: Advection	Transient storage	Inflow ($C_L = 0$)	Dispersion	Analytical solution; S_w linkage	Model real-world system
I	√				Eq. 7; Eq. 8	Straight, lined canal
II	√	√			Eq. 10; Eq. 12	Stream with uniform flow, very low dispersion
III	√	√	√		Eq. 14; Eq. 16	Stream with non-uniform flow,[a] very low dispersion
IV	√	√	√	√	Eq. 14; Eq. 17	Stream with non-uniform flow[a] and dispersion

[a]Inflow waters do not contain the added nutrient; C_L=0 (e.g. when radio-labeled nutrients are used).

cases: (1) nutrient concentrations have obtained a plateau such that steady-state conditions govern; (2) longitudinal dispersion is negligible; and (3) model parameters A, A_S, α, λ, and λ_S are spatially constant.

In Case I, stream discharge is uniform (spatially constant), solutes are transported by advection only (mixing due to dispersion and transient storage does not occur), and nutrient uptake occurs only in the main channel. This plug-flow system is best approximated in the real world by a straight, lined canal in which mixing is negligible. Given the conditions of uniform discharge, no transient storage, and the assumptions germane to all four cases, Eq. 4 becomes

$$\frac{Q}{A}\frac{dC}{dx} = -\lambda C \tag{6}$$

with an analytical solution given by

$$C_x = C_0 e^{-\frac{\lambda A}{Q}x} \tag{7}$$

Inspection of the analytical solution (Eq. 7) and the S_w equation (Eq. 2, with Q_x/Q_0 = 1.0 for uniform flow) provides an expression for uptake length as a function of the Case I transport model parameters:

$$S_w^I = \frac{Q}{A}(\lambda)^{-1} = \frac{velocity}{\lambda} \tag{8}$$

In Case II, stream discharge remains uniform, solutes are transported by advection and allowed to mix in accordance with the transient storage mechanism, and nutrient uptake occurs in the main channel and the transient storage zone. This conceptual system is best approximated in the real world by a natural stream with negligible mixing due to dispersion and a negligible change in flow over the length of the study reach. Given the conditions of uniform discharge and the assumptions germane to all four cases, Eq. 4 becomes

$$\frac{Q}{A}\frac{dC}{dx} = \alpha(C_S - C) - \lambda C \tag{9}$$

with an analytical solution given by

$$C_x = C_0 e^{-\frac{\Gamma}{Q}x} \tag{10}$$

where

$$\Gamma = \alpha A + \lambda A - \frac{\alpha^2 A^2}{\alpha A + \lambda_S A_S} \tag{11}$$

Inspection of the analytical solution (Eq. 10) and the S_w equation (Eq. 2) provides an expression for uptake length as a function of the Case II transport model parameters:

$$S_w^{II} = \frac{Q}{A}\left(\lambda + \frac{\alpha\lambda_S A_S}{\alpha A + \lambda_S A_S}\right)^{-1} \tag{12}$$

In Case III, stream discharge increases linearly with distance ($Q_x = Q_0 + q_L x$; non-uniform flow), solutes are transported by advection and allowed to mix in accordance with the transient storage mechanism, and nutrient uptake occurs in the main channel and the transient storage zone. This conceptual system is best approximated in the real world by a natural stream with negligible mixing due to dispersion and an increase in discharge over the length of the study reach. Given the assumption that the lateral inflow rate and concentration (q_L and C_L) are spatially constant, and the assumptions germane to all four cases, Eq. 4 becomes

$$\frac{Q}{A}\frac{dC}{dx} = \frac{q_L}{A}(C_L - C) + \alpha(C_S - C) - \lambda C \tag{13}$$

with an analytical solution given by

$$C_x = \frac{q_L C_L + (\theta C_0 - q_L C_L) e^{-\frac{\theta}{q_L}\ln\left(\frac{Q_x}{Q_0}\right)}}{\theta} \tag{14}$$

where

$$\theta = q_L + \alpha A + \lambda A - \frac{\alpha^2 A^2}{\alpha A + \lambda_S A_S} \tag{15}$$

To simplify the analysis, a modified version of Eq. 14 for the case of C_L equal to zero is linked to the S_w equation. This simplification corresponds to experimental additions in which the nutrient solution contains radio-labeled nutrients (e.g., containing ^{32}P; Newbold et al. 1981) or situations in which the transport equations are used to simulate background-corrected nutrient concentrations. Given this simplification, linkage of the analytical solution and the S_w equation provides an expression for uptake length as a function of the Case III transport model parameters:

$$S_w^{III} = \frac{Q_x - Q_0}{A}\left(\ln\frac{Q_x}{Q_0}\right)^{-1}\left(\lambda + \frac{\alpha\lambda_S A_S}{\alpha A + \lambda_S A_S}\right)^{-1} \tag{16}$$

Equations for Cases I to III above are developed using analytical solutions that ignore the dispersion term in Eq. 4. This simplification is a mathematical convenience that allows for the development of generic equations linking S_w and the transport model parameters (Eqs. 8, 12, and 16). Although this simplification is commonly made when analyzing steady-state

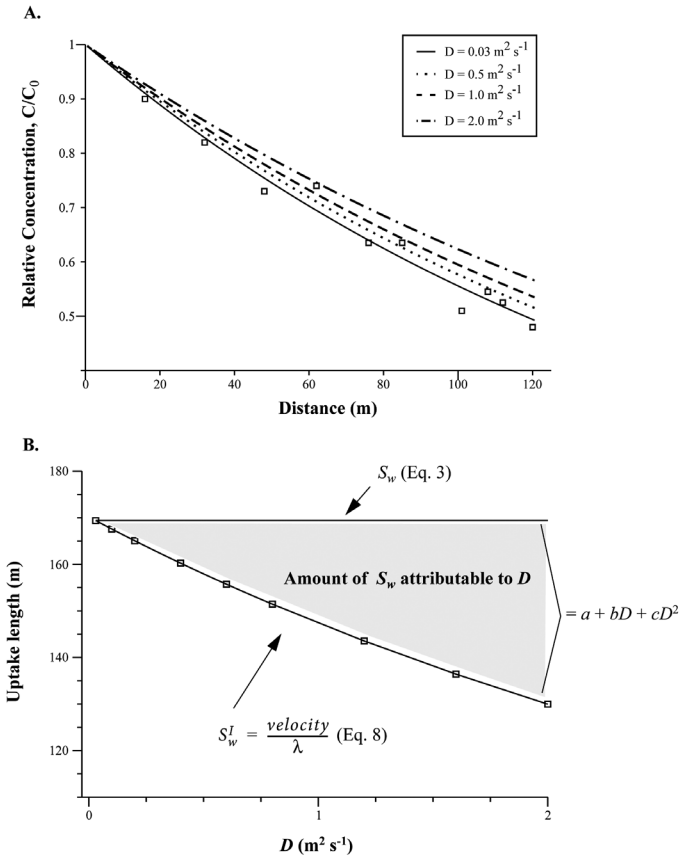


Fig. 1. Effect of dispersion on simulated concentrations (A) and estimates (B) of S_w.

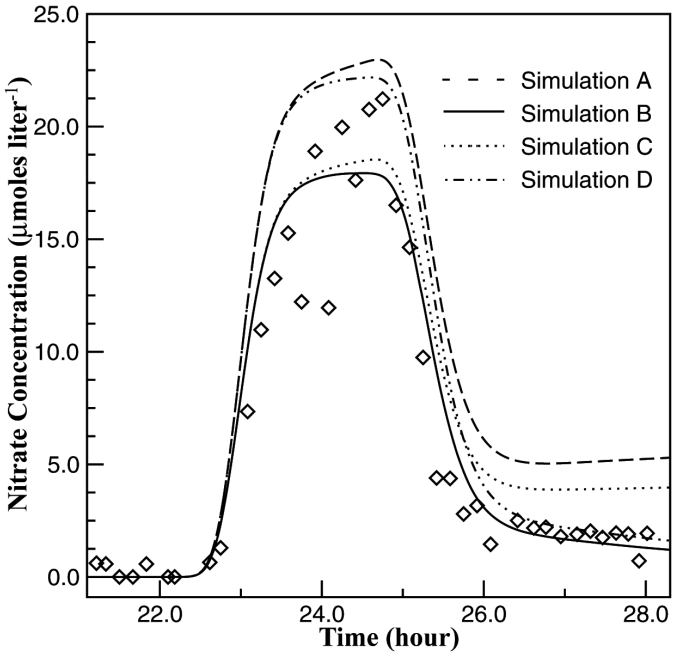


Fig. 2. Simulations of nitrate transport used in mass uptake calculations for the determination of main channel and storage zone uptake, Reach 2, Green Creek, Antarctica.

problems, it is important to note that the dispersion term may be of importance in many applications. As shown in Eq. 4, the dispersion term in the main channel equation includes the product of the dispersion coefficient and the concentration gradient, $\partial C/\partial x$. The dispersion term will therefore affect steady-state concentrations whenever a concentration gradient is present (dispersion causes mass to move downstream, thus influencing the time available for uptake). For the problems considered here, nutrient uptake in the main channel and/or storage zone results in a decrease in concentration with distance ($\partial C/\partial x < 0$), such that the effects of dispersion cannot always be ignored. These effects are illustrated in Figure 1A, where a transport model was used to simulate steady-state data (based on Figure 2 of Newbold et al. 1981). In the simulations, stream discharge ($Q_0 = Q_x = 3.8$ liters s^{-1}) and cross-sectional area ($A = 0.10$ m^2) were fixed using information provided in Newbold et al. (1981). An initial simulation with a dispersion coefficient of 0.03 m^2 s^{-1} used nonlinear regression (OTIS-P, Runkel 1998) to estimate the main channel uptake coefficient ($\lambda = 2.24 \times 10^{-4}$). Given this estimate of λ, three additional simulations were performed with higher values of the dispersion coefficient (D = 0.5, 1.0, and 2.0 m^2 s^{-1}). As shown in Figure 1A, increases in D lead to increases in concentration. (As D increases, mass disperses downstream more readily, such that main channel residence time is decreased. Because λ is a kinetic parameter, decreases in residence time result in less uptake and higher steady-state concentrations.)

In Case IV, an empirical equation linking S_w and the transport model parameters is developed for a stream in which dispersion is important. To quantify the effect of dispersion on S_w, a series of transport model simulations were performed using data and parameters corresponding to Case I (steady-state data and model parameters discussed above; Newbold et al. 1981). For each simulation, a hypothetical dispersion coefficient (D) was specified and nonlinear regression was used to estimate the main channel uptake coefficient (λ); the estimated uptake coefficient was in turn used to calculate S_w^I (= velocity/λ; Eq. 8). An estimate of S_w was also obtained using the steady-state data and linear regression (Eq. 3). The resulting values of S_w^I are plotted along with the estimate of S_w in Figure 1B. As shown in the figure, values of S_w^I depart from the linear regression estimate of S_w as the dispersion coefficient is increased. This departure results from the fact that higher values of λ are needed to produce the same longitudinal decline in nutrient concentration when higher values of D are specified (higher values of D result in reduced uptake, given a fixed value of λ; Figure 1A). The departure thus indicates that a portion of the S_w estimate from linear regression is attributable to dispersion (shaded area, Figure 1B). Similar analyses for Cases II and III indicate similar behavior, i.e., S_w^{II} and S_w^{III} from the transport model simulations depart from the linear regression estimate of S_w as values of the dispersion coefficient increase. Values of D less than

0.5 m^2 s^{-1} used in this analysis are generally representative of small streams affected by transient storage (e.g., 53 stream reaches in Runkel 2002), whereas values of D greater than 0.5 m^2 s^{-1} correspond to larger streams in which mixing is dominated by dispersion rather than transient storage.

An empirical relationship showing the effects of dispersion on S_w can be developed by considering the departure of S_w^I, S_w^{II}, and S_w^{III} from S_w as D increases. The departure ($S_w - S_w^{I,\ II,\ or\ III}$) is a curvilinear function of D that can be fit with a quadratic equation. Given this empirical relationship, a dispersion-corrected uptake length is given by

$$S_w^{IV} = \Phi + a + bD + cD^2 \quad (17)$$

where Φ is the right-hand side of Eq. 12 (Cases I and II) or Eq. 16 (Case III), and *a, b,* and *c* are coefficients of the quadratic equation. Note that Eq. 17 differs from the corresponding equations for Cases I to III, in that it represents an empirical result based on a single data set and specific parameter values. The corresponding equations for Cases I to III, in contrast, are general results that are applicable to any system under study.

Equations linking the S_w and transport-based approaches (Eqs. 8, 12, 16, and 17) each include a velocity and an uptake term. The presence of a velocity term is consistent with various studies that show a positive relationship between S_w and velocity (and/or discharge) (Valett et al. 1996, Peterson et al. 2001, Wollheim et al. 2001, Haggard et al. 2001). The uptake terms in Eqs. 12 and 16 include the sum of the main channel uptake coefficient (λ) and an effective storage zone uptake coefficient, given by

$$\lambda_S^{eff} = \frac{\alpha \lambda_S A_S}{\alpha A + \lambda_S A_S} \quad (18)$$

where λ_S^{eff} is known as the effective uptake coefficient because it accounts for the overall effect of storage zone uptake. The effective uptake coefficient corrects the observed uptake coefficient (λ_S) for two hydrologic effects of transient storage: (1) mass in the main channel must pass into the storage zone before being subject to removal by λ_S, and (2) the influence of λ_S on nutrient mass is a function of storage zone residence time. Values of λ_S^{eff} will often be less than λ_S because of the time required for nutrient mass to reach the storage zone; systems with large storage zones ($A_S > A$), however, will yield λ_S^{eff} in excess of λ_S provided the quantity $\alpha(1 - A/A_S)$ exceeds λ_S.

The effective storage zone uptake coefficient may be illustrated by revisiting the steady-state example (Figure 2 of Newbold et al. 1981), with $A_S = 0.05$ m^2, $D = 0.03$ m^2 s^{-1}, and $\alpha = 4 \times 10^{-4}$ s^{-1} (these parameter values are for illustrative purposes only). Given these parameters, the steady-state data were fit using nonlinear regression assuming uptake occurred exclusively in the main channel ($\lambda_S = 0$), yielding an estimate of the main channel uptake coefficient ($\lambda = 2.24 \times 10^{-4}$ s^{-1}). A second fit under the assumption of exclusive storage zone uptake ($\lambda = 0$) yields an estimate of the storage zone uptake coefficient ($\lambda_S = 1.02 \times 10^{-3}$ s^{-1}). Application of Eq. 18 for this second fit provides an effective uptake coefficient that is equal to that of the uptake coefficient provided by the initial main channel fit ($\lambda_S^{eff} = \lambda = 2.24 \times 10^{-4}$ s^{-1}). Equivalent amounts of mass removal are thus provided under two very different scenarios (main channel uptake with $\lambda > 0$ and $\lambda_S = 0$; storage zone uptake with $\lambda_S > 0$ and $\lambda = 0$). Additional simulations indicate that an optimal fit to the observed steady-state data may be obtained using any combination of λ and λ_S, provided the aggregate amount of uptake ($\lambda + \lambda_S^{eff}$) equals 2.24×10^{-4} s^{-1}.

Critique of the S_w approach

Several aspects of the S_w approach are discussed in this section in light of the linkage between the S_w and transport-based approaches presented above. Although this linkage is based on the model given by Eq. 4 and Eq. 5, it should be noted that other types of solute transport models could be used to make the same arguments (e.g., models with Michaelis-Menten uptake kinetics; Mulholland and DeAngelis 2000).

Interpretation of S_w—Uptake length is theoretically defined as the ratio of nutrient supply (flux) and use (uptake) (Newbold et al. 1981, 1983). Using this definition, S_w has been equated with the simple ratio of stream velocity and temporal uptake (S_w = velocity/λ; Newbold et al. 1981, 1983, Stream Solute Workshop 1990). This equality has subsequently been used to convert between temporal and spatial uptake coefficients (Mulholland et al. 1985, Butturini and Sabater 1998) and in the development of uptake velocity (see "Hydrologic effects on uptake velocity"). Use of this equality should be viewed with caution, however, as the relationship holds only for the very restrictive assumptions of Case I (no transient storage, inflow, or dispersion; Eq. 8). Natural systems vary in complexity, but few (if any) correspond directly to Case I. As shown by Cases II to IV (Eqs. 12, 16, and 17), estimates of S_w incorporate the effects several hydrologic processes (advection, dispersion, inflow, transient storage; Table 1), such that use of S_w = velocity/λ is inappropriate for most systems.

S_w as a measure of aggregate uptake—Investigators have long acknowledged the physical and biogeochemical importance of transient storage in determining nutrient fate (Valett et al. 1996, Mulholland et al. 1997, Valett et al. 1997, Grimaldi and Chaplot 2000, Harvey and Wagner 2000, Fellows et al. 2001, Hall et al. 2002, Runkel et al. 2003, Sheibley et al. 2003, Thomas et al. 2003, Bohlke et al. 2004, Gooseff et al. 2004, Gücker and Boëchat 2004, McKnight et al. 2004, Mulholland et al. 2004, Ensign and Doyle 2005). Despite this knowledge, the primary modeling tool for the analysis of nutrient uptake is the calculation of S_w, a quantity whose theoretical basis can be linked to the solution of an advection-dispersion equation with a single first-order uptake term (Eq. 2). Estimates of S_w thus represent an aggregate measure of uptake that does not distinguish between main channel and storage zone processes (i.e., main channel and storage zone processes are represented using single quantity, S_w). Use of the S_w approach may therefore obscure important process information by lumping different uptake processes into a single

uptake term (Cases II-IV; Eqs. 12, 16, and 17). Analyses based on uptake length may therefore be of limited use given the observation that storage zone uptake processes often differ dramatically from benthic processes occurring within the main channel (Mulholland et al. 1997).

The aggregation of main channel and storage zone uptake represents a limitation of the S_w approach that is not easily remedied. As per the earlier discussion of the effective storage zone rate coefficient (Eq. 18), steady-state data do not provide enough information for the estimation of both main channel and storage zone uptake (i.e., fitting λ and λ_S to steady-state data using nonlinear regression results in a non-unique solution). Other techniques, such as the analysis of time-variable data, are therefore needed to go beyond the aggregate measure of uptake given by S_w (see "A transport-based approach").

Process integration and S_w comparisons—As noted earlier, an advantage of the S_w approach is the ability to integrate the suite of hydrologic and nonhydrologic processes affecting nutrient fate into a single metric. In regard to hydrologic processes, estimates of S_w can be influenced by the hydrologic processes of advection, dispersion, inflow, and transient storage (Eq. 17). This hydrologic influence stems from the derivation of the S_w equation, where a simple solution to the advection-dispersion equation ignores the effects of dispersion, transient storage, and inflow (Stream Solute Workshop 1990; Case I). As such, these processes become part of the S_w estimate (Eq. 17, Case IV) in most natural systems, where mixing processes and/or non-uniform flow conditions affect nutrient fate (Cases II-IV; e.g., 53 stream reaches in Runkel 2002). In addition to this hydrologic influence, estimates of S_w represent an aggregate measure of main channel and storage zone uptake (see "S_w as a measure of aggregate uptake").

Given the complexity inherent in natural systems and the resultant effects of hydrologic and nonhydrologic processes, quantitative interpretation of S_w when performing inter- and intrastream comparisons may be a difficult task. Specifically, an "apples and oranges" problem may arise when comparing S_w estimates from individual stream reaches and/or different nutrients. The meaning of S_w for two stream reaches, for example, may be very different if one of the reaches has uniform flow and negligible dispersion (Case II, Eq. 12) whereas the second has non-uniform flow and nonnegligible dispersion (Case IV, Eq. 17). Internutrient comparisons may also be confounded if uptake of one nutrient is dominated by main channel processes (e.g., benthic uptake of phosphate, approximated by Case I) whereas the other is influenced by both main channel and storage zone uptake (e.g., benthic uptake and hyporheic denitrification, approximated by Case II). One of the strengths of the S_w approach, the ability to integrate both transport and uptake into single metric, thus appears to be a disadvantage when applied to natural systems that are inherently different. Further, most natural streams are best approximated by Case IV, such that numerous parameters are lumped into S_w. An infinite number of parameter combinations can therefore produce a given value of S_w, leading to conflicting interpretations of uptake length.

Hydrologic effects on uptake velocity—As noted above, the effects of hydrologic and nonhydrologic processes have led to difficulties in interpretation when estimates of S_w are compared among streams with differing hydrologic conditions. Alternative metrics of uptake have therefore been developed in an attempt to isolate the effects of biogeochemical processes (Stream Solute Workshop 1990, Davis and Minshall 1999, Hall et al. 2002). One such metric, uptake velocity, is given by

$$v_f = \frac{velocity \times depth}{S_w} \tag{19}$$

while a second metric, the areal uptake rate (U), is the simple product of v_f and nutrient concentration. These calculations of v_f and U as a function of S_w are routinely made when analyzing data from experimental additions (Marti et al. 1997, Davis and Minshall 1999, Tank et al. 2000, Dodds et al. 2000, Sabater et al. 2000, Mulholland et al. 2000, Peterson et al. 2001, Dodds et al. 2002, Hall et al. 2002, Hall and Tank 2003, Mulholland et al. 2004).

As given by Eq. 19, v_f is thought to normalize S_w for the effects of varying velocity. As such, v_f is preferred over S_w as a metric for temporal and spatial comparisons of nutrient dynamics (Davis and Minshall 1999). The development of v_f as a parameter that is free of hydrologic effects appears to be predicated on the idea that S_w is equal to velocity/λ (Case I, Eq. 8; see "Interpretation of S_w"). Given this definition of S_w, Eq. 19 becomes

$$v_f = \lambda \times depth \tag{20}$$

where v_f is a quantity that is independent of velocity and other hydrologic factors. Unfortunately the relationship given by Eq. 20 only holds when S_w is calculated for a system corresponding to Case I, a situation that is approximated by a straight, lined canal in which mixing is negligible. Simple, plug-flow canal systems are rarely the systems of interest when studying nutrient dynamics, and estimates of v_f based on S_w (Eq. 19) are thus rarely the simple product shown in Eq. 20. This can be seen by considering the form of Eq. 19 as complexity increases from the simple canal system (Case I) to natural systems in which transient storage, inflow, and dispersion affect mass transport (Case IV). At the highest level of complexity, S_w is given by Eq. 17, and Eq. 19 becomes

$$v_f = \frac{velocity \times depth}{\frac{Q_x - Q_0}{A}\left(\ln\frac{Q_x}{Q_0}\right)^{-1}\left(\lambda + \frac{\alpha\lambda_S A_S}{\alpha A + \lambda_S A_S}\right)^{-1} + a + bD + cD^2} \tag{21}$$

such that v_f is no longer independent of hydrologic factors. The dependence of v_f on hydrologic factors may explain the difficulties in interpretation that are noted in the literature (e.g., Hall et al. 2002). Because U is the product of v_f and concentration, hydrologic factors may also lead to difficulties in interpretation when comparing estimates of areal uptake rate. In summary, the hydrologic processes that are implicitly part of the S_w estimate (transient storage, inflow, dispersion) also

Table 2. Comparison of S_w and transport-based approaches

	S_w approach	Transport approach	Error, %
Regression coefficient (Eq. 3; m^{-1})	0.0336	—	—
First-order uptake coefficient (λ; s^{-1})	—	2.42×10^{-3}	—
Uptake length (S_w; m)	29.8	26.3	13.0
Uptake velocity (v_f; m s^{-1})	6.63×10^{-5}	7.49×10^{-5}	–11.5
Areal uptake rate (U; μg P m^{-2} h^{-1})	524.9	593.3	–11.5

affect estimates of v_f and U. This effect of hydrologic processes on v_f and U is contrary to the conventional thought that these metrics are not a function of the stream system's hydrologic characteristics (e.g., Sabater et al. 2000, Doyle et al. 2003).

A transport-based approach

In light of the critique presented above, two applications of a new transport-based approach are presented. The first application illustrates the transport-based approach for the steady-state data sets that have been collected by researchers to date. The second application illustrates the approach when time-series data are available. Both applications employ a solute transport model that includes the effects of transient storage. The purpose of most of the model simulations is to estimate one or more parameters describing hydrologic transport and/or nutrient uptake. Solute transport models that provide an objective means of parameter estimation are therefore preferable (e.g., nonlinear regression as in OTIS-P, Runkel 1998, http://co.water.usgs.gov/otis).

Steady-state data sets—Although it may be preferable to study nutrient uptake using time-series data, most existing data sets are of the steady-state variety. An approach to analyzing existing data sets is therefore presented here, in hopes of addressing some of the limitations of the S_w approach when only steady-state data are available. Although the approach is based on analysis of steady-state nutrient data, it is assumed that existing data sets include time-series data for a co-injected conservative tracer. The approach consists of 3 steps:

(1) Using time-series data for the co-injected conservative tracer and a transient storage solute transport model (e.g., OTIS-P), estimate the parameters governing conservative transport (A, A_s, D, α). When conducting this step, it is important to distinguish between the effects of dispersion and transient storage on mixing of the conservative tracer; mixing caused by the dispersive process can have a significant effect on the analysis of steady-state nutrient data (Figure 1).

(2) Using steady-state nutrient data, the transport parameters from step 1, and a transient storage solute transport model (e.g., OTIS-P), estimate the aggregate amount of nutrient uptake using the main channel uptake coefficient, λ. The steady-state data needed for this step consist of observed nutrient concentrations collected at two or more locations after the conservative tracer has reached a steady-state plateau. As discussed earlier, steady-state data do not provide enough information for the separate estimation of main channel and storage zone uptake. The approach presented here for steady-state data sets therefore does not address a critical limitation of the S_w approach (i.e., inability to distinguish between main channel and storage zone uptake). As with the S_w approach, the aggregate effect of main channel and storage zone uptake is considered using a single parameter (λ).

(3) Using the estimate of aggregate uptake from step 2 (λ), calculate uptake length [$S_w = Q/(A\lambda)$], uptake velocity ($v_f = \lambda \times$ depth), and areal uptake rate ($U = v_f \times$ concentration). Calculation of v_f and U in this manner (as a function of λ rather than S_w) results in parameters that are not a function of hydrologic processes (see text surrounding Eq. 20).

Parameter estimation in Steps 1 (A, A_s, D, α) and 2 (λ) may be facilitated through the use of automated techniques such as nonlinear regression (e.g., OTIS-P, Runkel 1998).

The transport-based approach for steady-state data is illustrated using the data and transport analysis presented by Mulholland et al. (1997). Phosphate data presented in Figures 4 and 5 of Mulholland et al. (1997) may be used to obtain estimates of S_w, v_f, and U under the S_w approach (Eqs. 3 and 19; Table 2). Estimates of these three metrics may also be obtained by implementing steps 2 and 3 above (Table 2), using the reported transport parameters ($A = 0.087$ m^2, $A_s = 0.129$ m^2, $D = 0.2$ m^2 s^{-1}, $Q_0 = 4.9 \times 10^{-3}$ m^3 s^{-1}, $q_L = 1.63 \times 10^{-5}$ m^3 s^{-1} m^{-1}, $\alpha = 4 \times 10^{-4}$ s^{-1}). Discrepancies between the two sets of estimates are attributable to the hydrologic processes of dispersion, transient storage, and inflow. The inability of the S_w approach to explicitly consider these processes results in considerable error (Table 2) that may confound interpretation of S_w, v_f, and U. The transport-based approach is therefore recommended for the analysis of steady-state data sets.

Time-series data sets—The transport-based analysis for steady-state data sets presented above addresses several limitations of the S_w approach, but does not address the inability of the S_w approach to distinguish between main channel and storage zone uptake. Due to the aforementioned difficulties in estimating both main channel and storage zone uptake from steady-state data, this final limitation is best addressed through the analysis of time-series data. A transport-based approach for time-series data is illustrated using the data and transport analysis first presented by McKnight et al. (2004). The work of McKnight et al. (2004) is revisited to provide addi-

Table 3. Mass uptake calculations used to determine the percent uptake in main channel and storage zone, using data and simulations from Green Creek, Antarctica.

	A	B	C	D	E	F	G	H	I	J
	Mass passing reach endpoint, moles (Eq. 22)				Total mass uptake	Initial estimate of uptake, %			Final estimate of uptake, %	
Reach	Sim. A	Sim. B	Sim. C	Sim. D	(A – B), moles	M.C. $\left(\frac{A-C}{E}\right)$	S.Z. $\left(\frac{A-D}{E}\right)$	Sum (F + G), %	M.C. (100 × F /H)	S.Z. (100 × G /H)
1	1.285	0.931	0.945	1.263	0.354	95.9	6.1	101.9	94.0	6.0
2	0.435	0.327	0.350	0.405	0.108	78.3	27.6	105.9	73.9	26.1
3	0.179	0.060	0.060	0.179	0.120	99.9	0.4	100.3	99.6	0.4

tional details on the approach, and to correct some small errors in the previous analysis for nitrate uptake in reaches 2 to 3 of the modeled system. As described by McKnight et al. (2004), an experimental nitrate addition was conducted in Green Creek, Antarctica, on January 13, 1995. A time series of nitrate concentration was collected at the terminus of four stream reaches, located 50, 226, 327, and 497 meters downstream of the injection site. Data from the first three reaches are used to illustrate the transport-based approach for time-series data, consisting of the following four steps:

(1) Using time-series data for the co-injected conservative tracer and a transient storage solute transport model, estimate the parameters governing conservative transport (see step 1 of "Steady-state data sets").
(2) Using time-series nutrient data, the transport parameters from step 1, and a transient storage solute transport model, estimate main channel and storage zone uptake (λ and λ_s) in each reach.
(3) Conduct four simulations for each reach to quantify the percentage of mass loss occurring in the main channel and storage zone. The four simulations are as follows:

Simulation A (conservative transport): simulate conservative transport in the current reach by setting λ and λ_s equal to zero (current reach and all downstream reaches); λ and λ_s in all upstream reaches remain at the values estimated in step 2.

Simulation B (reactive transport): simulate reactive transport in the main channel and storage zone of the current reach by leaving λ and λ_s at the values estimated in step 2 (current reach and all upstream reaches); λ and λ_s in all downstream reaches are set to zero.

Simulation C (reactive transport, main channel uptake only): simulate reactive transport in the main channel of the current reach by leaving λ at the value from step 2 and setting λ_s equal to zero (current reach). Uptake coefficients (λ and λ_s) for all upstream reaches remain at the values estimated in step 2; λ and λ_s in all downstream reaches are set to zero.

Simulation D (reactive transport, storage zone uptake only): simulate reactive transport in the storage zone of the current reach by leaving λ_s at the value from step 2 and setting λ equal to zero (current reach). Uptake coefficients (λ and λ_s) for all upstream reaches remain at the values estimated in step 2; λ and λ_s in all downstream reaches are set to zero.

Results of simulations A–D for reach 2 of Green Creek are shown in Figure 2. Differences between simulation A and simulation B reflect the total amount of uptake occurring within reach 2. Similarly, differences between simulation A and simulation C reflect main channel uptake, whereas differences between simulation A and simulation D reflect storage zone uptake (Figure 2). These differences may be formally quantified by determining the mass passing the reach endpoint for each simulation. The mass is determined by integrating the area under the nutrient load (stream discharge times concentration, *QC*) versus time curve at the reach endpoint, as given by:

$$mass = cf\int_0^t Q_x(C_x - C_b)dt \qquad (22)$$

where C_x is the simulated nutrient concentration, C_b is the background nutrient concentration, Q_x is stream discharge, and t is time. The value of the conversion factor, cf, is dependent on the units associated with Q and C, and the desired units for *mass*. For Green Creek, background nitrate concentrations are negligible ($C_b = 0$), *mass* is in moles of nitrate, and cf equals 3.6 (C_x has units of micromoles nitrate liter^{-1}, Q_x has units of m^3 s^{-1}, t is in hours).

Applying Eq. 22 to the results provided by simulations A–D allows for the calculation of total mass uptake and a partitioning of the total uptake between the main channel and storage zone. The amount of uptake occurring in the main channel and storage zone, expressed as a percentage of total uptake, is given by

$$\%uptake_{mainchannel} = \frac{\text{main channel uptake}}{\text{total uptake}} = \frac{mass_{simA} - mass_{simC}}{mass_{simA} - mass_{simB}} \qquad (23)$$

$$\%uptake_{storagezone} = \frac{\text{storage zone uptake}}{\text{total uptake}} = \frac{mass_{simA} - mass_{simD}}{mass_{simA} - mass_{simB}} \qquad (24)$$

where the *mass* terms are developed using the integration from Eq. 22.

Because simulations C and D overestimate uptake, Eqs. 23 and 24 overestimate the uptake percentage (i.e., for simulation C, $\lambda_S = 0$ results in a higher main channel concentration relative to simulation B and more main-channel uptake; for simulation D, $\lambda = 0$ results in a higher storage zone concentration and more storage zone uptake). The percentages calculated using Eqs. 23 and 24 therefore need to be adjusted proportionately downward such that the sum is equal to 100%.

Quantification of mass uptake using simulations A–D and Eqs. 22–24 is illustrated for Green Creek in Table 3. Total mass uptake is first calculated using the results of simulations A and B (Eq. 22; column E, Table 3). This quantity is used in Eqs. 23 and 24 to develop initial estimates of mass uptake in the main channel and storage zone (columns G and H, Table 3). The sum of the mass uptake percentages (column H, Table 3) is then used to arrive at the final estimates of uptake (columns I and J, Table 3). These final estimates indicate that the bulk of the uptake (~74% to 100%) occurred within the main channel, where benthic uptake by dense algal mats is likely (McKnight et al. 2004). Substantial uptake (~26%) also occurred in the storage zone of reach 2, where uptake is attributed to microbial processes (McKnight et al. 2004). As shown by this example, the transport-based approach allows for the quantification of mass uptake on a reach-by-reach basis, as well as a determination of where the uptake is taking place in a given reach (main channel versus storage zone uptake).

(4) Calculate estimates of uptake length, uptake velocity, and areal uptake rate. Estimates of S_w, v_f, and U such as those developed using steady-state data may also be developed for the case of time-series data. This task is accomplished by analyzing a synthetic, steady-state data set. The synthetic data set is generated by performing a steady-state simulation using the uptake coefficients (λ and λ_S) developed in step 2. This steady-state simulation produces a synthetic nutrient data set that may be used to develop estimates of S_w, v_f, and U (i.e., using output from the steady-state simulation as data, repeat the calculations described in steps 2 and 3 for "Steady-State data sets"). Estimation of S_w, v_f, and U in this manner does not address the inability of the S_w approach to distinguish between main channel and storage zone uptake, a limitation that is addressed primarily through the calculations presented in step 3 (Eqs. 23 and 24). An additional means of addressing this limitation is the development of alternate, location-specific estimates of uptake velocity and areal uptake rate. For example, the uptake coefficients developed in step 2 may be used to estimate uptake velocities for the main channel and storage zone:

$$v_f^{mc} = \lambda d \tag{25}$$

$$v_f^{sz} = \lambda_S d_S \tag{26}$$

where d is the average main channel depth, d_S is the average storage zone depth, v_f^{mc} is the main channel uptake velocity, and v_f^{sz} is the storage zone uptake velocity. Average depths in Eqs. 25 and 26 may be calculated using estimates of cross-sectional area (A, A_S) and the appropriate width terms (average main channel width and average lateral extent of the storage zone). Due to the hydrologic effects of transient storage noted earlier, main channel and storage zone uptake velocities may not be directly compared for a given reach (e.g., the bulk of uptake in Green Creek occurs in the main channel, yet $v_f^{mc} < v_f^{sz}$). Despite this limitation, v_f^{mc} and v_f^{sz} may prove to be useful metrics for interreach and interstream comparisons. Analogous measures of areal uptake rate (U^{mc}, U^{sz}) may be developed using the results of Eqs. 25 and 26 and the appropriate nutrient concentrations.

Summary and Conclusions

In this article, a critical review of the S_w approach is presented, based on an analysis of the equations describing solute transport and uptake in streams. This analysis is used to point out several limitations of the S_w approach that arise from (1) the method of data collection (steady-state data), and (2) the method of data analysis (S_w equation and linear regression). As a result of these limitations, future research should focus on the collection of time-series data and subsequent data analysis using the transport-based approach presented here. This recommendation is based on four aspects of the transport-based approach summarized below.

Quantification of uptake processes—The rates and types of uptake processes occurring in the main channel and storage zone may differ dramatically due to differences in pH, temperature, redox condition, and the concentration of dissolved gases (Mulholland et al. 1997, Runkel et al. 2003). Differentiation between main channel and storage zone uptake is therefore of paramount importance when studying uptake in a given system. The transport-based approach for time-series data explicitly considers both main channel and storage zone uptake and allows for the estimation of rates for both areas. This approach to analyzing experimental data is thus consistent with the body of work on the physical effects of transient storage and numerous studies that attribute a portion of the observed uptake to storage zone processes.

Process separation—A key aspect of the transport-based approach is the ability to separately quantify the effects of hydrologic processes (as represented by a suite of physical parameters) and nonhydrologic processes (biotic uptake and sorption, as represented by λ and λ_S). The ability to objectively separate processes leads to estimates of first-order uptake coefficients (λ and λ_S) that are independent of hydrology. Estimates of v_f and U obtained using the transport-based approach are also independent of hydrologic effects, a feature of the approach that may lead more consistent interstream comparisons. The use of first-order uptake coefficients also provides consistency with standard methods for quantifying kinetic processes. Standard methods for considering the effect of tem-

perature on reaction kinetics, for example, may be directly applicable to the estimates of λ and λ_S (e.g., use of published temperature correction constants for first-order reactions; Brown and Barnwell 1987, Chapra 1997).

Field considerations—Under the transport-based approach for time-series data, the nutrient addition need not continue until nutrient concentrations reach steady-state plateau concentrations. As noted by several authors, excessively long duration nutrient additions can lead to errors in uptake estimates due to the potential for regeneration and mineralization (Davis and Minshall 1999, Mulholland et al. 2000, Payn et al. 2005). Use of time-series data allows for short-term, nonplateau injections which minimize these sources of error and increase the likelihood of measuring the desired process, gross uptake. This feature of the transport-based approach may in turn allow researchers to investigate longer stream reaches without introducing the effects of regeneration and mineralization. Another field aspect of the transport-based approach is the collection of nutrient samples at the reach endpoint. A comparison of the S_w approach and the transport-based analysis suggests that the transport-based analysis requires more nutrient data than the S_w approach. Although this may be true, the additional information gained from the time-series approach (the ability to quantify main channel and storage zone uptake) appears to warrant the extra effort. Further, the amount of extra effort required may be minimal if one considers the work of Dodds et al. (2002), where 20 or more samples are recommended for the S_w approach.

Analysis considerations—The transport-based analysis for Green Creek is a multireach problem in which unsteady flow is considered. Analysis of most nutrient addition experiments will be significantly simpler than the Green Creek example because of two factors. First, many addition experiments are conducted over a short length of stream (50 to 300 m), such that analysis of data from a single reach endpoint is adequate. Second, most additions occur when flow is nominally steady over the sampling period. Given a steady flow regime, the discharge term in Eq. 22 may be moved outside the integral. This allows for the integration of the time-versus-concentration curve, information that is directly available as output from most solute transport models (e.g., OTIS output files include a time series of concentrations at specific user-defined locations). Finally, it should be noted that many researchers investigating nutrient uptake are already employing solute transport models to quantify the physical process of transient storage. Analyses of nutrient uptake under the transport-based approach should therefore be relatively straightforward, as the required expertise is comparable to that needed for the analysis of transient storage.

In summary, the transport-based, time-series approach to quantifying nutrient uptake provides a means of estimating traditional measures of nutrient uptake (S_w, v_f, U) while providing additional information on the location and magnitude of uptake (main channel versus storage zone). This additional information may provide further insight into the processes controlling nutrient uptake and transport in natural systems.

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